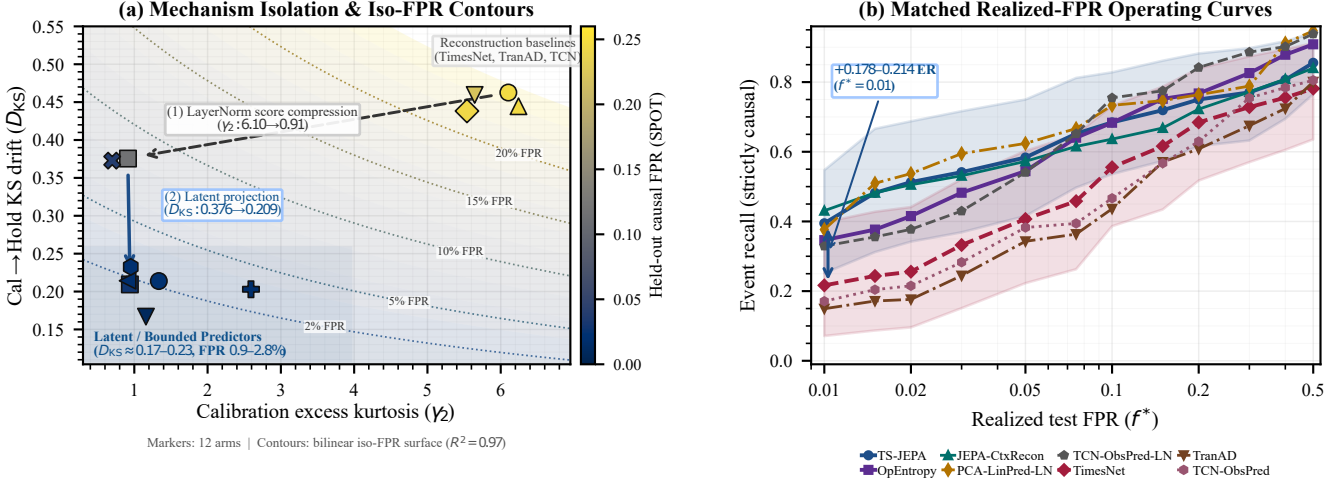

(a) Mechanism Isolation & Iso-FPR Contours
Reconstruction baselines (TimesNet, TranAD, TCN)
(1) LayerNorm score compression (γ_2: 6.10→0.91)
(2) Latent projection (D_{KS}: 0.376→0.209)
20% FPR
15% FPR
10% FPR
5% FPR
2% FPR
Latent / Bounded Predictors (D_{KS} ≈ 0.17–0.23, FPR 0.9–2.8%)
Cal→Hold KS drift (D_{KS})
Calibration excess kurtosis (γ_2)
Held-out causal FPR (SPOT)
Markers: 12 arms | Contours: bilinear iso-FPR surface (R^2 = 0.97)
(b) Matched Realized-FPR Operating Curves
+0.178–0.214 ER (f^* = 0.01)
Event recall (strictly causal)
Realized test FPR (f^*)
TS-JEPA
JEPA-CtxRecon
TCN-ObsPred-LN
TranAD
OpEntropy
PCA-LinPred-LN
TimesNet
TCN-ObsPred